JavaScript Module 1 - Chapter 2: How JavaScript Works Internally

Introduction

JavaScript does not execute directly. A JavaScript engine reads the source code, parses it, optimizes it, and executes it. Modern browsers use highly optimized engines to make JavaScript fast.

JavaScript Engine

A JavaScript engine is software that executes JavaScript code. Examples include Google's V8, Mozilla's SpiderMonkey, Apple's JavaScriptCore, and Chakra (legacy Microsoft engine).

Execution Process

1. 1. Read the source code.
2. 2. Tokenization and Parsing.
3. 3. Generate an Abstract Syntax Tree (AST).
4. 4. Convert to bytecode or optimized machine code using JIT compilation.
5. 5. Execute the program.

Parser

The parser checks syntax and converts source code into an Abstract Syntax Tree (AST). Syntax errors stop execution before the program runs.

JIT Compilation

Modern engines use Just-In-Time (JIT) compilation. Frequently executed code is optimized into machine code for better performance.

Memory

JavaScript automatically allocates memory and removes unused objects through Garbage Collection.

Browser vs Node.js

In browsers, JavaScript interacts with the DOM and Web APIs. In Node.js, it interacts with the operating system, files, network, and servers.

Interview Questions

1. What is a JavaScript engine?
2. What is V8?
3. What is an AST?
4. Explain JIT compilation.
5. Difference between browser runtime and Node.js runtime.

Practice

Draw the JavaScript execution flow: Source Code → Parser → AST → JIT Compiler → Machine Code → Execution.